

```
MODEL_NAME = 'ssd_mobilenet_v1_coco_11_06_2017'
```

```
MODEL_FILE = MODEL_NAME + '.tar.gz'
```

```
DOWNLOAD_BASE = 'http://download.tensorflow.org/models/object\_detection/'
```

Path to frozen detection graph. This is the actual model that is used for the object detection.

```
PATH_TO_CKPT
= MODEL_NAME + '/
frozen_inference_graph.pb'
```

List of the strings that are used to add a correct label for each box.

```
PATH_TO_LABELS
= os.path.join('data',
'mscoco_label_map.pbtxt')
```



```
NUM_CLASSES = 90
```

Programming Environment For Object Detection

```
opener = urllib.request.URLopener()
opener.retrieve(DOWNLOAD_BASE + MODEL_FILE, MODEL_FILE)
tar_file = tarfile.open(MODEL_FILE)
for file in tar_file.getmembers():
    file_name = os.path.basename(file.name)
    if 'frozen_inference_graph.pb' in file_name:
        tar_file.extract(file, os.getcwd())
```

```
detection_graph = tf.Graph()
with detection_graph.as_default():
    od_graph_def = tf.GraphDef()
    with tf.gfile.GFile(PATH_TO_CKPT, 'rb') as fid:
        serialized_graph = fid.read()
        od_graph_def.ParseFromString(serialized_graph)
        tf.import_graph_def(od_graph_def, name='')
```

```
label_map = label_map_util.load_labelmap(PATH_TO_LABELS)
categories = label_map_util.convert_label_map_to_categories(label_map,
```